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(54) **VACUUM SEALING LID AND VACUUM SEALING DEVICE**

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(57)

ABSTRACT

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A vacuum sealing lid comprises a first lid, a second lid, a first sealing ring and a second sealing ring; the first lid is used to fit a first size of a Mason jar, a first placement groove is arranged on a top of the first lid, and a first vent hole is arranged in a middle of the first placement groove; the second lid is used to fit a second size of a Mason jar, a second placement groove is arranged on a top of the second lid, and a second vent hole is arranged in a middle of the second placement groove, the second lid is detachably connected with the first lid; the first sealing ring is assembled on the first lid.

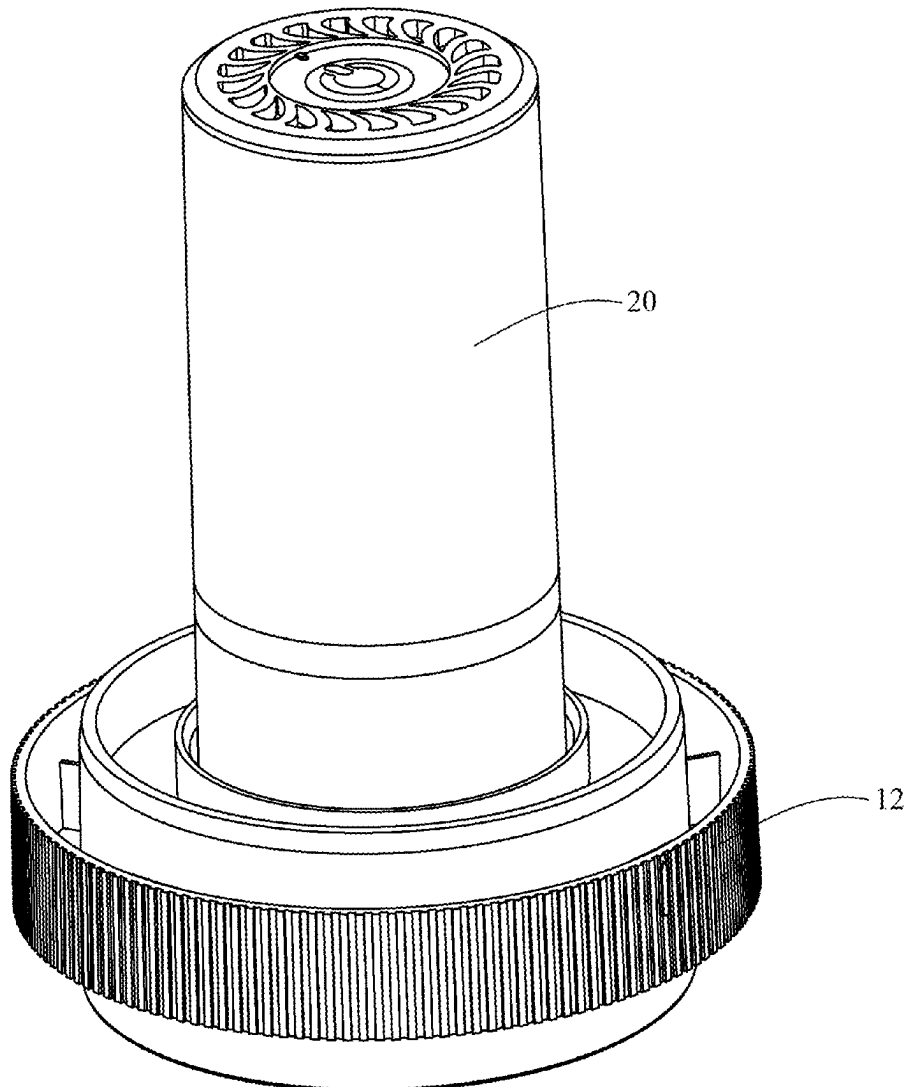
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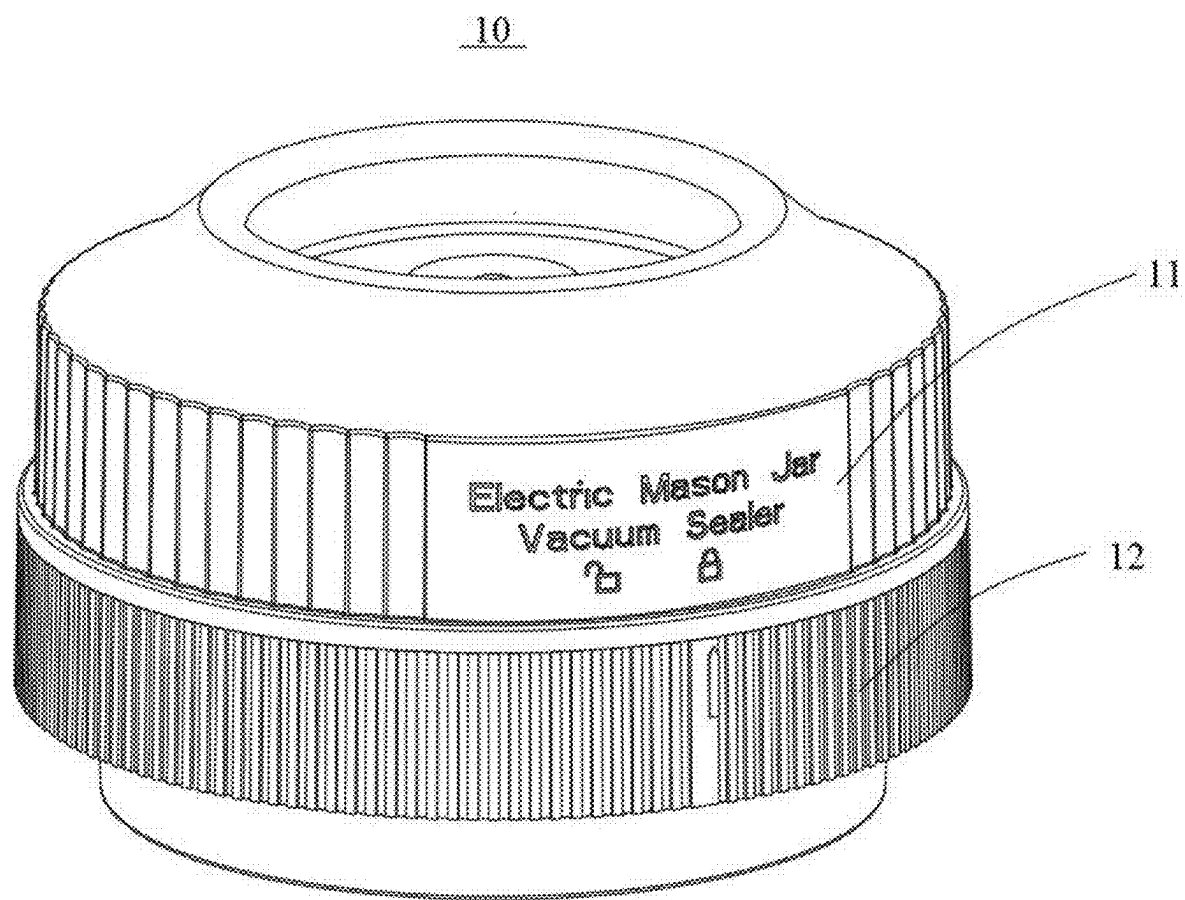


FIG. 1

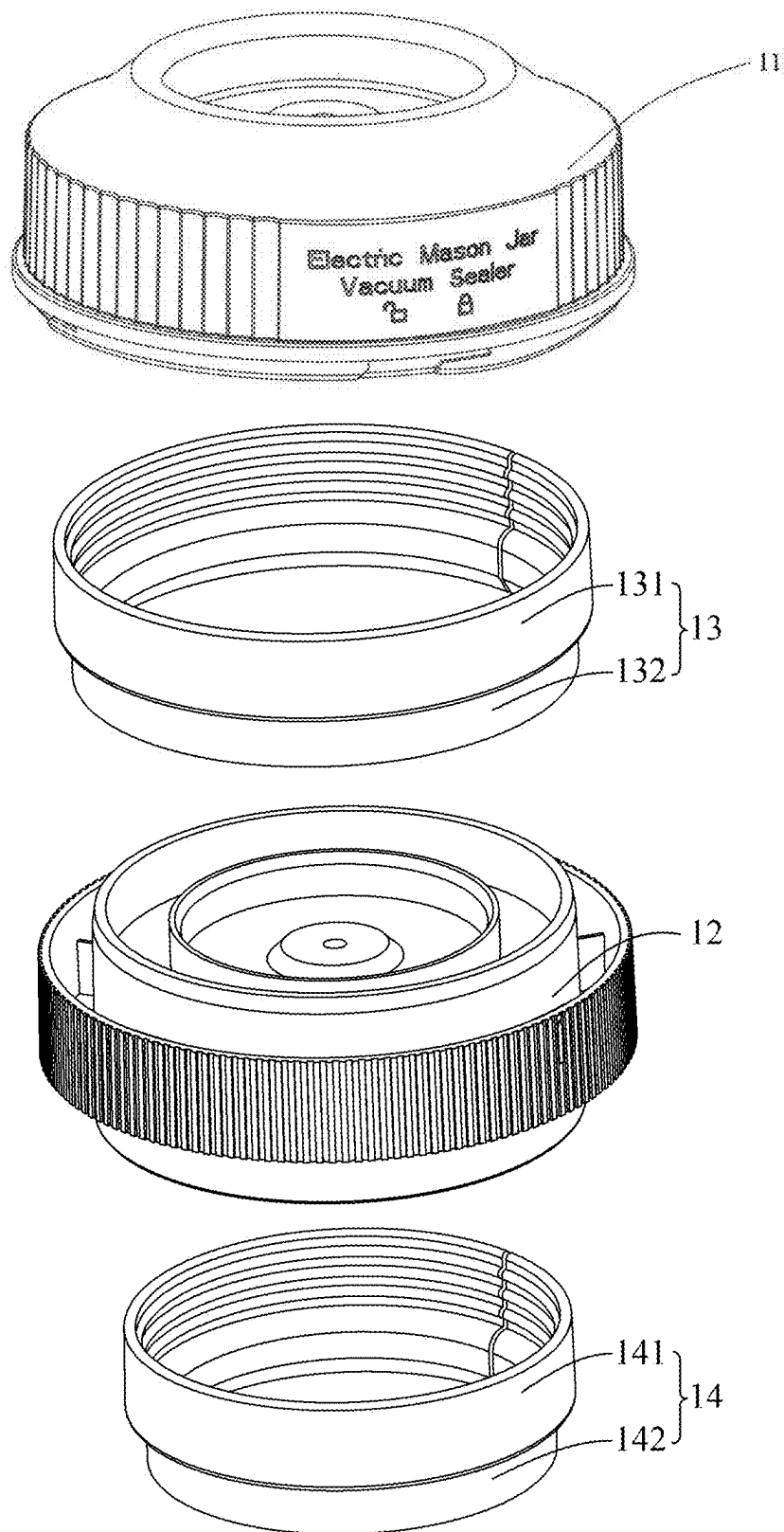


FIG. 2

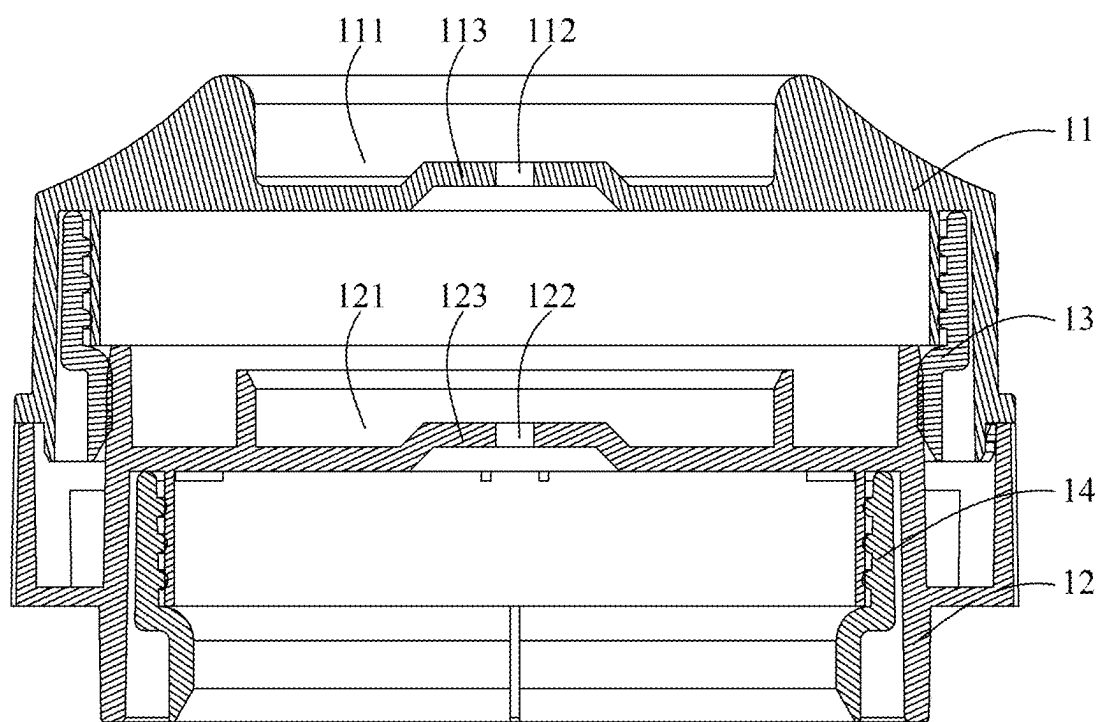


FIG. 3

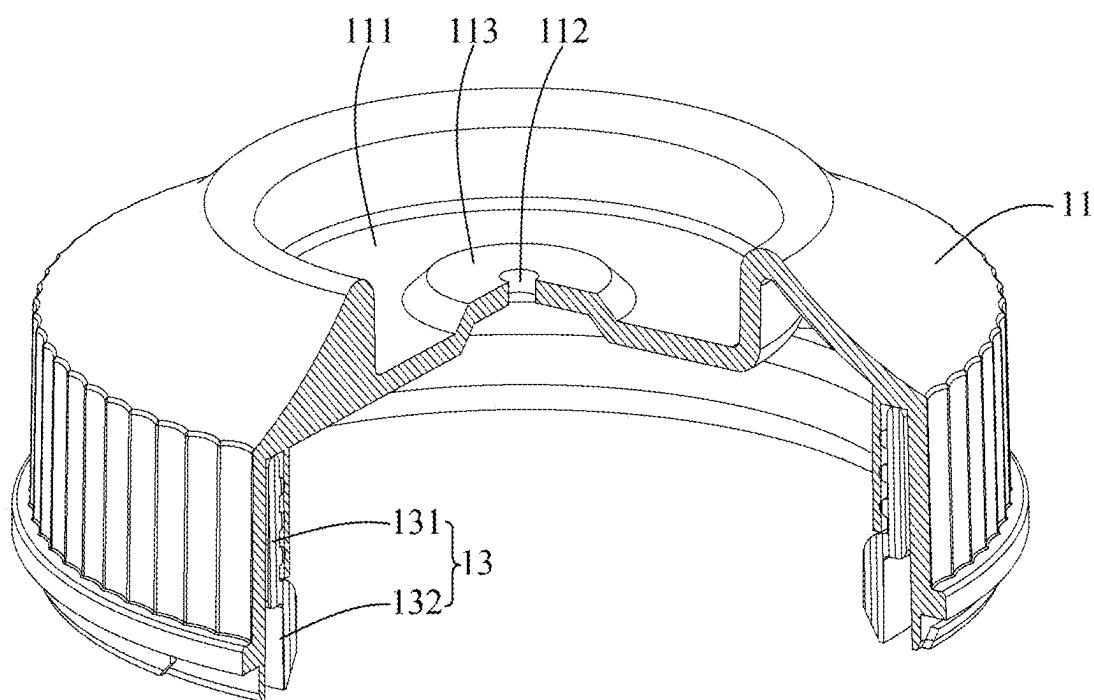


FIG. 4

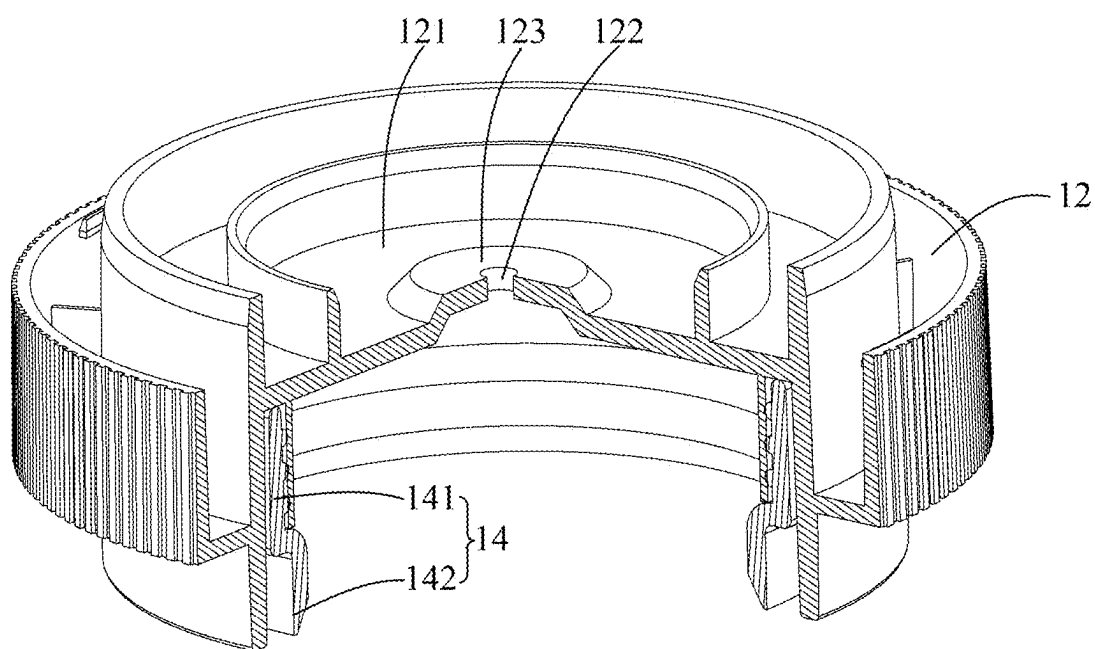


FIG. 5

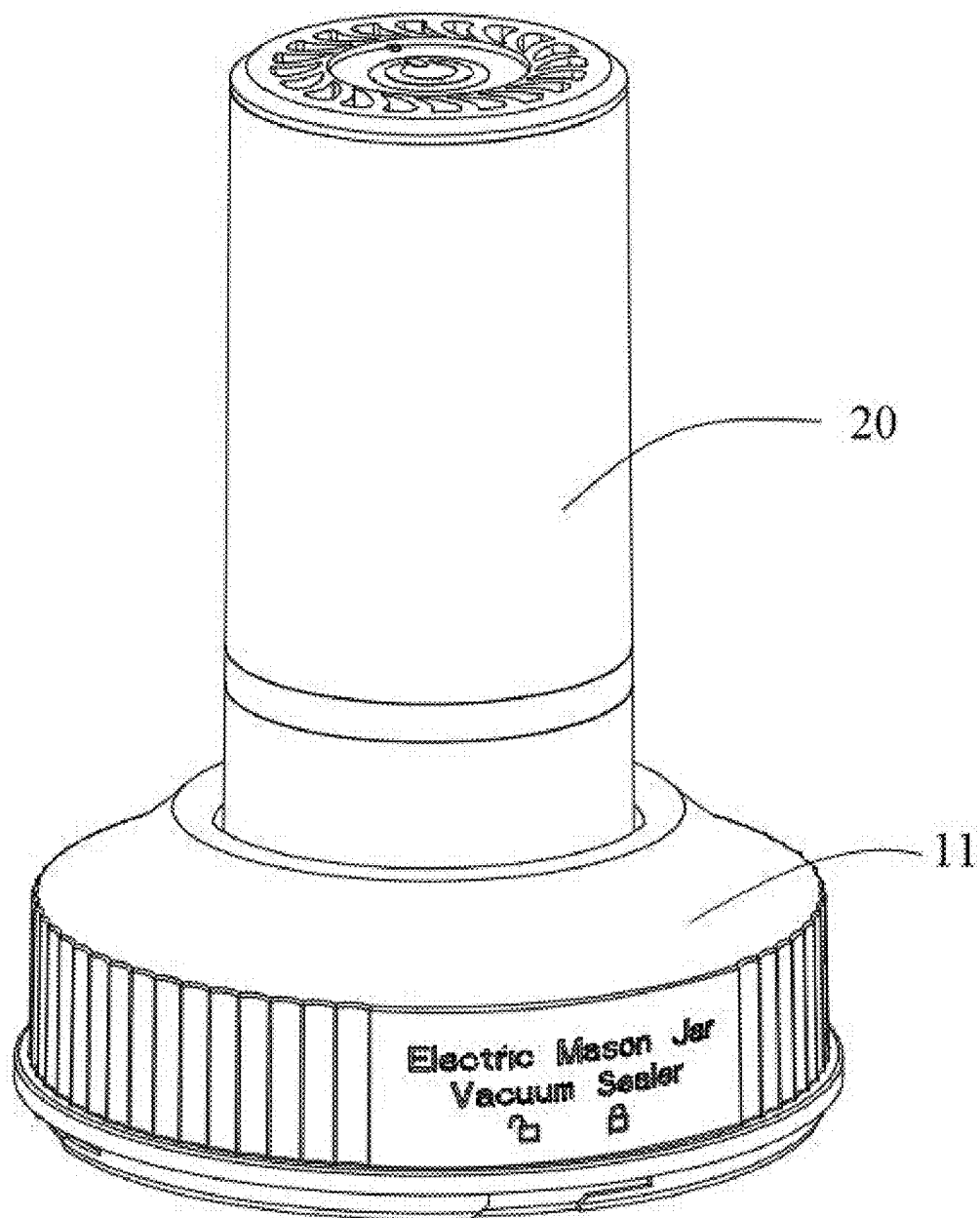


FIG. 6

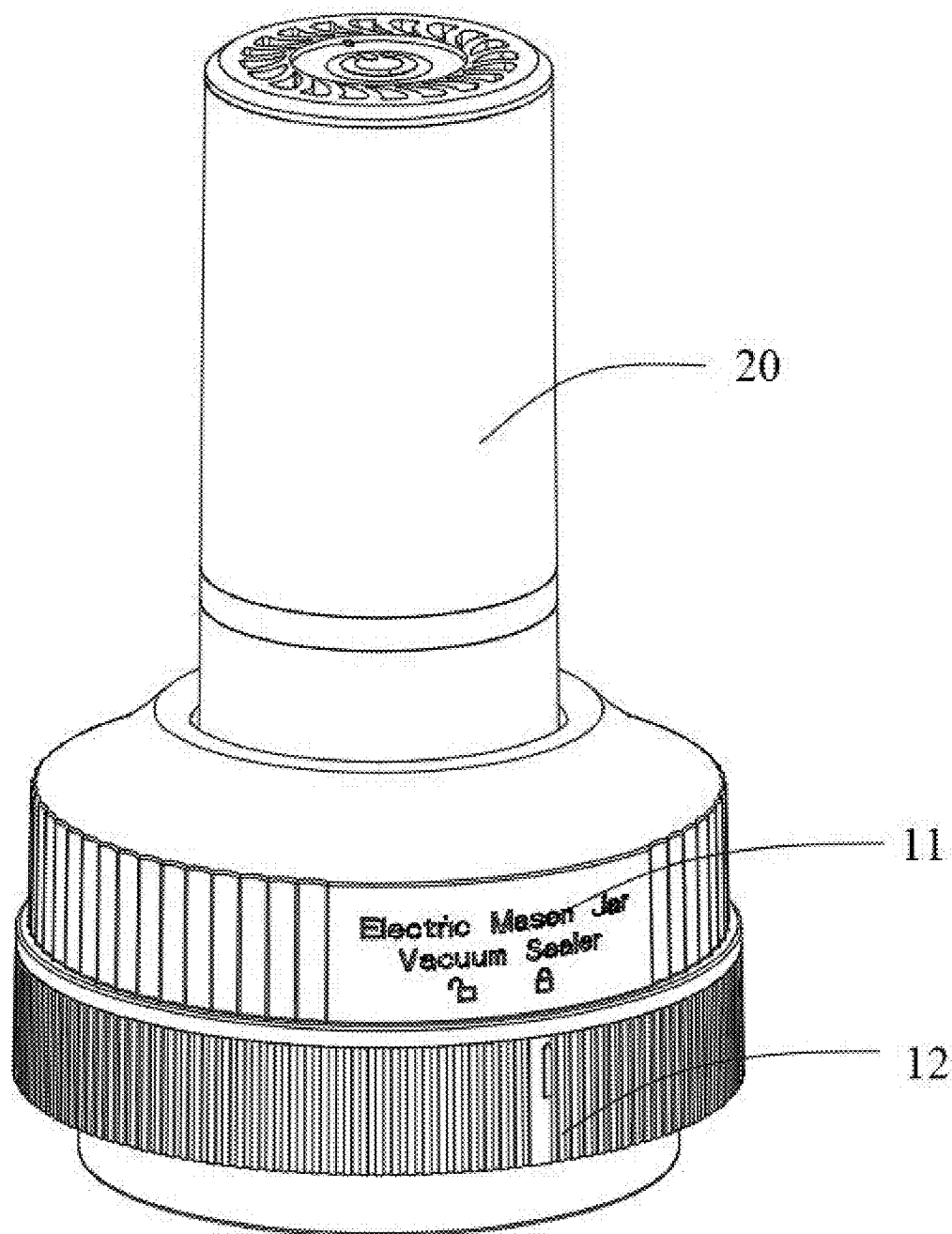


FIG. 7

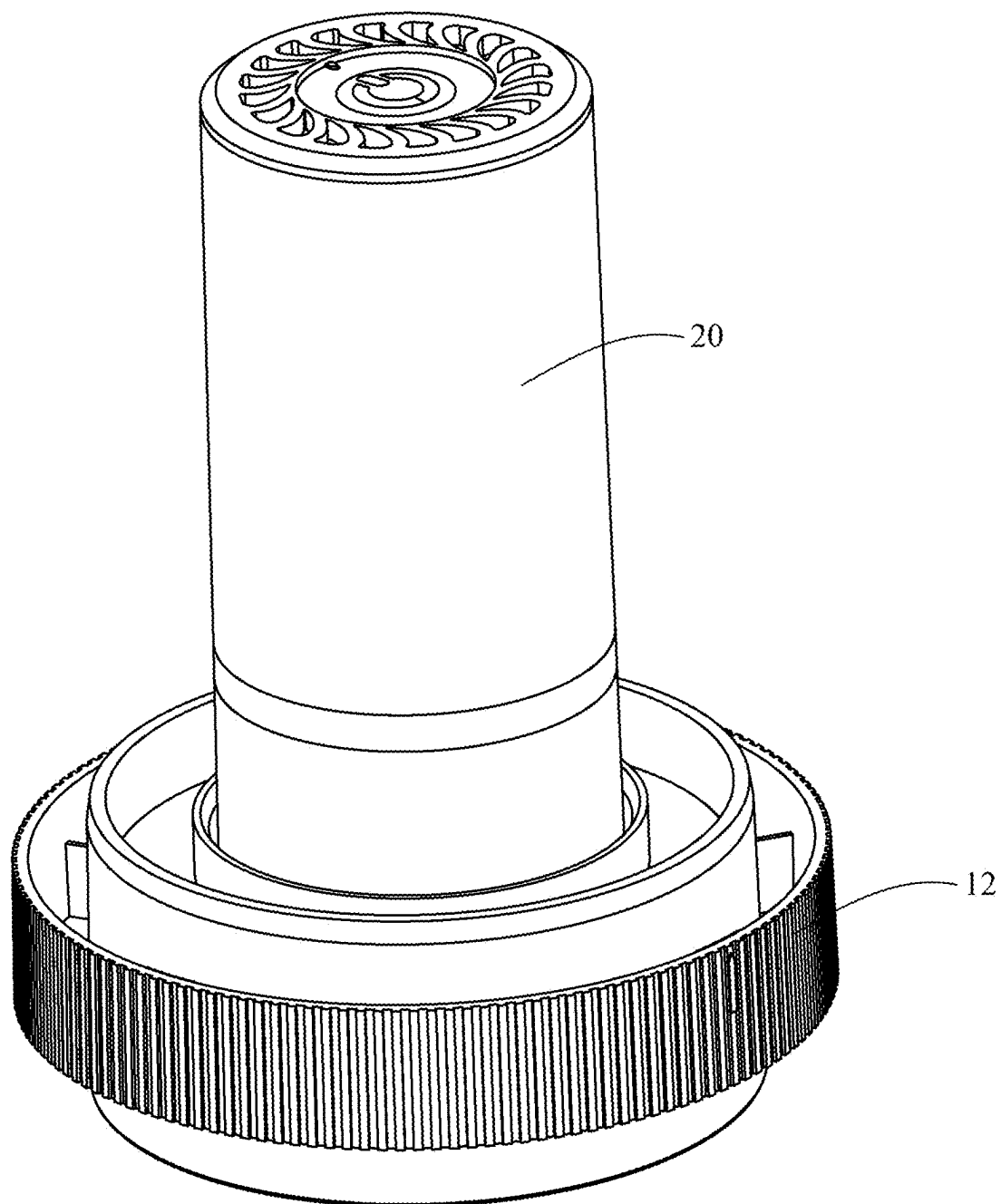


FIG. 8

VACUUM SEALING LID AND VACUUM SEALING DEVICE

TECHNICAL FIELD

[0001] The invention relates to the technical field of Mason jar sealing devices, in particular to a vacuum sealing lid and a vacuum sealing device.

BACKGROUND ART

[0002] A Mason jar is a type of glass jar with a threaded iron lid, available in two sizes. During use, an external vacuum sealing device is used to vacuumize the Mason jar, so that the iron lid is tightly attached to a top opening of the glass jar, achieving a sealed effect.

[0003] The Patent with the publication number of CN219668956 U describes a vacuum sealing device, comprising an upper shell and a lower shell that are interlock and fixed to each other, an installation chamber is formed between the upper shell and the lower shell, vacuum components for vacuumizing are installed inside the installation chamber, a bottom of the lower shell is provided with two circular sealing soft adhesives of different sizes to accommodate Mason jars of different sizes. Currently, the vacuum sealing device integrates the sealing conversion component and vacuum components into one unit, which results in a large volume and inconvenience in storage; moreover, since the vacuum components inside the shell cannot be used separately, the additional purchase of a vacuum machine for vacuumizing other containers or vacuum bags is required, resulting in high costs.

SUMMARY OF THE INVENTION

[0004] The main object of the invention is to provide a vacuum sealing lid, which aims to provide a vacuum sealing lid with simple structure, convenient storage, and separate design from the vacuum machine.

[0005] To achieve the above object, the invention provides the following technical scheme: a vacuum sealing lid, applied to Mason jars and cooperating with a vacuum machine to realize vacuuming of Mason jars of different sizes, the vacuum sealing lid comprises a first lid, a second lid, a first sealing ring and a second sealing ring,

[0006] the first lid is used to fit a first size of a Mason jar, a first placement groove is arranged on a top of the first lid, and a first vent hole is arranged in a middle of the first placement groove,

[0007] the second lid is used to fit a second size of a Mason jar, a second placement groove is arranged on a top of the second lid, and a second vent hole is arranged in a middle of the second placement groove, the second lid is detachably connected with the first lid,

[0008] the first sealing ring is assembled on the first lid, the second lid and the first lid are sealed and connected through the first sealing ring to form a vacuum chamber, both the first vent hole and the second vent hole are connected to the vacuum chamber,

[0009] the second sealing ring is assembled on the second lid.

[0010] Optionally, the top of the first lid is recessed inward to form the first placement groove, and a top of the first placement groove is provided with a fillet or chamfer.

[0011] Optionally, a first boss is arranged in the middle of the first placement groove for positioning the vacuum machine, and the first vent hole is located on the first boss.

[0012] Optionally, the top of the second lid is protrusively provided with a shielding ring, so as to form the second placement groove, an inner side of a top of the second placement groove is provided with a fillet or chamfer.

[0013] Optionally, a second boss is arranged on the top of the second lid for positioning the vacuum machine, the second boss is located inside the shielding ring, the second vent hole is located on the second boss.

[0014] Optionally, a first annular positioning groove is arranged on one side of the first lid facing the second lid, the first sealing ring comprises a first clamping portion and a first sealing portion connected to each other, the first clamping portion is clamped in the first annular positioning groove, and the first sealing portion is tightly attached to the second lid or the first size of the Mason jar.

[0015] Optionally, a second annular positioning groove is arranged on one side of the second lid facing the Mason jar, the second sealing ring comprises a second clamping portion and a second sealing portion connected to each other, the second clamping portion is clamped in the second annular positioning groove, and the second sealing portion is tightly attached to the second size of the Mason jar.

[0016] Optionally, the detachable connection between the first lid and the second lid is achieved through rotary buckles.

[0017] Optionally, both outer sides of the first lid and the second lid are provided with anti-slip grooves.

[0018] The invention also proposes a vacuum sealing device, the vacuum sealing device comprises the vacuum sealing lid and the vacuum machine,

[0019] the vacuum sealing lid is as described above,

[0020] the vacuum machine cooperates with either the first placement groove or the second placement groove to vacuumize the Mason jars.

[0021] The technical scheme of the invention comprises the first lid, the second lid, the first sealing ring and the second sealing ring, the first lid and the second lid are respectively used to fit the Mason jars of different sizes, the tops of the first lid and the second lid are provided with placement grooves for positioning the vacuum machine, the first sealing ring is assembled on the first lid, the second sealing ring is assembled on the second lid, the lids achieve the tight fit with the Mason jars through the sealing rings; at the same time, the sealed connection is achieved between the second lid and the first lid through the first sealing ring. The scheme adopts the technical approach of detachably connecting two lids and cooperating with the vacuum machine to vacuumize different sizes of Mason jars, it effectively solves the technical problems of large size, inconvenient storage, and high cost in the prior art. As a result, it achieves the technical effects of simple structure, convenient storage, and separate design from the vacuum machine.

BRIEF DESCRIPTION OF ACCOMPANY DRAWINGS

[0022] FIG. 1 is a schematic diagram of the structure of one embodiment of the vacuum sealing lid of the invention;

[0023] FIG. 2 is a schematic diagram of the disassembled structure of the vacuum sealing lid of the invention;

[0024] FIG. 3 is a schematic diagram of the cross-section of the vacuum sealing lid of the invention;

[0025] FIG. 4 is a schematic diagram of the connection structure of the first lid and the first sealing ring of the invention;

[0026] FIG. 5 is a schematic diagram of the connection structure of the second lid and the second sealing ring of the invention;

[0027] FIG. 6 is a schematic diagram of the structure of one embodiment of the vacuum sealing device of the invention;

[0028] FIG. 7 is a schematic diagram of the structure of another embodiment of the vacuum sealing device of the invention;

[0029] FIG. 8 is a schematic diagram of the structure of yet another embodiment of the vacuum sealing device of the invention.

ATTACHED DINGS MARKS

[0030] 100 vacuum sealing device; 10 vacuum sealing lid; 11 first lid; 111 first placement groove; 112 first vent hole; 113 first boss; 12 second lid; 121 second placement groove; 122 second vent hole; 123 second boss; 13 first sealing ring; 131 first clamping portion; 132 first sealing portion; 14 second sealing ring; 141 second clamping portion; 142 second sealing portion; 20 vacuum machine.

Specific Embodiment of the Invention

[0031] The technical schemes in the embodiments of the invention will be clearly and completely described in combination with the accompanying drawings in the embodiments of the invention. Obviously, the described embodiments are only some of the embodiments of the invention, but not all of the embodiments. Based on the embodiments in this invention, all other embodiments obtained by those of ordinary skill in the art without making creative efforts shall fall within the protection scope of this invention.

[0032] It should be noted that all directional indications (such as up, down, left, right, front, back, etc.) in the embodiments of the invention are only used to explain the relative position relationship and movement of the components under a certain posture (as shown in the figures). If the certain posture changes, the directional indication will also change accordingly.

[0033] In the invention, it should also be noted that the terms of “connect”, “fix”, etc., should be generally understood unless there are specific restrictions or stipulations, for example, the “connect” may refer to fixed connection, detachable connection or integral connection; the “connect” may also refer to mechanical connection or electrical connection; the means of “connect” may be directly connected or indirectly connected through an intermediate medium, and may be internal communication between the two elements. For those skilled in the art, the specific meaning of the above terms in the invention can be understood according to the specific situation.

[0034] In addition, in the invention, the terms “first” and “second” are only used for descriptive purposes, and should not be understood as indicating or implying relative importance or implying the number of indicated technical features. Therefore, the features defined with “first” or “second” may expressly or implicitly include one or more of the features. In addition, the technical schemes between the various embodiments can be combined with each other, but it must

be based on the fact that those of ordinary skill in the art can implement it. When the combination of technical schemes is contradictory or cannot be implemented, it should be considered that such combination of technical schemes does not exist and is not within the protection scope claimed by the invention.

[0035] The invention proposes a vacuum sealing lid 10, the vacuum sealing lid 10 is applied to Mason jars and cooperates with a vacuum machine 20 to realize vacuuming of Mason jars of different sizes.

[0036] In order to better understand the above technical scheme, the above technical scheme will be described in detail below in combination with the accompanying drawings and specific embodiments.

[0037] In the embodiment of the invention, as shown in FIG. 1, FIG. 2 and FIG. 3, the vacuum sealing lid 10 comprises a first lid 11, a second lid 12, a first sealing ring 13 and a second sealing ring 14, wherein the first lid 11 is used to fit a first size of a Mason jar, a first placement groove 111 is arranged on a top of the first lid 11, and a first vent hole 112 is arranged in a middle of the first placement groove 111; the second lid 12 is used to fit a second size of a Mason jar, a second placement groove 121 is arranged on a top of the second lid 12, and a second vent hole 122 is arranged in a middle of the second placement groove 121, the second lid 12 is detachably connected with the first lid 11, the first sealing ring 13 is assembled on the first lid 11, the second lid 12 and the first lid 11 are sealed and connected through the first sealing ring 13 to form a vacuum chamber, both the first vent hole 112 and the second vent hole 122 are connected to the vacuum chamber; the second sealing ring 14 is assembled on the second lid 12.

[0038] It is understandable that Mason jars come in two different sizes, and the two lids of the invention are designed to fit these two different sizes. Specifically, as shown in FIG. 4, the first lid 11 is used to fit the first size of the Mason jar, while as shown in FIG. 5, the second lid 12 is used to fit the second size of the Mason jar, the first lid 11 and the second lid 12 are connected in a detachable manner. When vacuumizing Mason jars of different sizes, the vacuum sealing lid 10 can be disassembled to select the corresponding lid. As shown in FIG. 3, the two lids are assembled in a partially overlapping manner, which makes the structure compact when stored and reduces the assembly height. Both lids are provided with sealing rings, the first lid 11 is assembled with the first sealing ring 13, and the second lid 12 is assembled with the second sealing ring 14, wherein the first sealing ring 13 has two functions; when the first lid 11 and the second lid 12 are assembled, the first sealing ring 13 is located between the two lids to achieve a sealing effect; when the first lid 11 and the second lid 12 are separated and the first lid 11 is matched with the first size of the Mason jar, the first sealing ring 13 is located between the first lid 11 and the Mason jar to achieve a sealing effect; when the second lid 12 is matched with the second size of the Mason jar, the second sealing ring 14 is located between the second lid 12 and the Mason jar to achieve a sealing effect.

[0039] Both the first lid 11 and the second lid 12 of the invention are provided with placement grooves for positioning the vacuum machine 20. As shown in FIG. 6, when vacuumizing the first size of the Mason jar, the first lid 11 and the second lid 12 are separated, and the first lid 11 achieves a tight fit with the first size of the Mason jar through the first sealing ring 13; at the same time, the vacuum

machine 20 is placed in the first placement groove 111, and the vacuum machine 20 is started, the air inside the glass jar is expelled through the first vent hole 112 until the metal lid is tightly attached to a top opening of the glass jar. When vacuumizing the second size of the Mason jar, there are two ways, that is, the first lid 11 and the second lid 12 can be in an assembled state or in a separated state. As shown in FIG. 7, when the first lid 11 and the second lid 12 are in the assembled state, the vacuum chamber is formed between the first lid 11 and the second lid 12 through the first sealing ring 13; the first vent hole 112 and the second vent hole 122 respectively communicate the vacuum chamber and aligned vertically; the second lid 12 achieves a tight fit with the second size of the Mason jar of through the second sealing ring 14; at the same time, the vacuum machine 20 is placed in the first placement groove 111, and the vacuum machine 20 is started, the air inside the glass jar enters the vacuum chamber through the second vent hole 122 and is expelled through the first vent hole 112 until the iron lid is tightly attached to a top opening of the glass jar. As shown in FIG. 8, when the first lid 11 and the second lid 12 are in the separated state, the second lid 12 achieves a tight fit with the second size of the Mason jar through the second sealing ring 14; at the same time, the vacuum machine 20 is placed in the second placement groove 121, and the vacuum machine 20 is started, the air inside the glass jar is expelled through the second vent hole 122 until the iron lid is tightly attached to the top opening of the glass jar.

[0040] The first lid 11 and the second lid 12 of the invention are detachably connected; meanwhile, the vacuum machine 20 is also designed to be detachable from the vacuum sealing lid 10, reducing storage volume and enhancing flexibility during use; the vacuum machine 20 can be used for vacuumizing Mason jars and can also be used for vacuumizing other containers or vacuum bags.

[0041] The technical scheme of the invention comprises the first lid 11, the second lid 12, the first sealing ring 13 and the second sealing ring 14, the first lid 11 and the second lid 12 are respectively used to fit the Mason jars of different sizes, the tops of the first lid 11 and the second lid 12 are provided with placement grooves for positioning the vacuum machine 20, the first sealing ring 13 is assembled on the first lid 11, the second sealing ring 14 is assembled on the second lid 12, the lids achieve the tight fit with the Mason jars through the sealing rings; at the same time, the sealed connection is achieved between the second lid 12 and the first lid 11 through the first sealing ring 13. The scheme adopts the technical approach of detachably connecting two lids and cooperating with the vacuum machine 20 to vacuumize different sizes of Mason jars, it effectively solves the technical problems of large size, inconvenient storage, and high cost in the prior art. As a result, it achieves the technical effects of simple structure, convenient storage, and separate design from the vacuum machine 20.

[0042] In the embodiment of the invention, as shown in FIG. 4, the top of the first lid 11 is recessed inward to form the first placement groove 111, and a top of the first placement groove 111 is provided with a fillet or chamfer. The inward recess of the first placement groove 111 can further reduce the height after assembly and decrease the storage volume. The setting of the fillet or chamfer serves as a guide for the placement of the vacuum machine 20.

[0043] In the embodiment of the invention, as shown in FIG. 4, a first boss 113 is arranged in the middle of the first

placement groove 111 for positioning the vacuum machine 20, and the first vent hole 112 is located on the first boss 113. A bottom of the vacuum machine 20 is provided with a positioning groove. When the vacuum machine 20 is placed in the first placement groove 111, positioning is achieved through the cooperation of the first boss 113 and the positioning groove.

[0044] In the embodiment of the invention, as shown in FIG. 5, the top of the second lid 12 is protrusively provided with a shielding ring, so as to form the second placement groove 121, an inner side of a top of the second placement groove 121 is provided with a fillet or chamfer. The setting of the fillet or chamfer on the top of the second placement groove 121 serves as a guide for the placement of the vacuum machine 20.

[0045] In the embodiment of the invention, as shown in FIG. 5, a second boss 123 is arranged on the top of the second lid 12 for positioning the vacuum machine 20, the second boss 123 is located inside the shielding ring, the second vent hole 122 is located on the second boss 123. The bottom of the vacuum machine 20 is provided with the positioning groove. When the vacuum machine 20 is placed in the second placement groove 121, positioning is achieved through the cooperation of the second boss 123 and the positioning groove.

[0046] In the embodiment of the invention, as shown in FIG. 4, a first annular positioning groove is arranged on one side of the first lid 11 facing the second lid 12, the first sealing ring 13 comprises a first clamping portion 131 and a first sealing portion 132 connected to each other, the first clamping portion 131 is clamped in the first annular positioning groove, and the first sealing portion 132 is tightly attached to the second lid 12 or the first size of the Mason jar.

[0047] In the embodiment of the invention, as shown in FIG. 5, a second annular positioning groove is arranged on one side of the second lid 12 facing the Mason jar, the second sealing ring 14 comprises a second clamping portion 141 and a second sealing portion 142 connected to each other, the second clamping portion 141 is clamped in the second annular positioning groove, and the second sealing portion 142 is tightly attached to the second size of the Mason jar.

[0048] In the invention, in order to ensure the secure assembly of the first sealing ring 13, multiple annular protrusions are arranged on an inner or outer side of the first clamping portion 131. These annular protrusions increase friction by contacting an inner surface of the first annular positioning groove, thereby preventing the first sealing ring 13 from loosening. Similarly, the second sealing ring 14 is also equipped with the same structure, serving the same object.

[0049] In the embodiment of the invention, the detachable connection between the first lid 11 and the second lid 12 is achieved through rotary buckles. The rotary buckles specifically includes clamping grooves and clamping buckles. In one embodiment of the invention, the clamping grooves are arranged on the first lid 11, four clamping grooves are provided and evenly distributed in a circular pattern; each clamping groove is L-shaped, and comprises a connected notch and a locking groove, wherein the notch is located at the lower position; the clamping buckles are arranged on the second lid 12, the number of clamping buckles and clamping grooves are the same and their positions correspond. During

the assembly process, the clamping buckles are aligned with the notches, the first lid **11** presses against the second lid **12** to allow the clamping buckles to enter the locking grooves through the notches; subsequently, the first lid **11** and the second lid **12** are rotated in opposite directions, causing the clamping buckles located within the locking grooves to slide towards one side away from the notches, thereby achieving the locking of the first lid **11** and the second lid **12**. In other embodiments of the invention, it is also possible to arrange the clamping grooves on the second lid **12** and the clamping buckles on the first lid **11**; furthermore, the number of clamping grooves and clamping buckles can be altered according to actual requirements.

[0050] In the invention, as shown in FIG. 1, the first lid **11** and the second lid **12** are also provided with markings and arrow indicators to indicate the locked or unlocked status, making it convenient for users to operate.

[0051] In the embodiment of the invention, as shown in FIG. 1, both outer sides of the first lid **11** and the second lid **12** are provided with anti-slip grooves. The setting of anti-slip grooves can increase friction, facilitating the assembly and disassembly of the first lid **11** and the second lid **12**.

[0052] The invention also proposes a vacuum sealing device **100**, as shown in FIG. 6, FIG. 7, and FIG. 8, the vacuum sealing device **100** comprises the vacuum sealing lid **10** and the vacuum machine **20**. The specific structure of the vacuum sealing lid **10** follows the embodiments described above. Since this vacuum sealing device **100** adopts all the technical schemes of the aforementioned embodiments, it possesses at least all the advantageous effects brought by these embodiments, which are not repeated herein. The vacuum machine **20** cooperates with either the first placement groove **111** or the second placement groove **121** to vacuumize Mason jars.

[0053] The above are only optional embodiments of the invention, and do not limit the patent scope of the invention. All equivalent structural changes made by using the specification of the invention and accompanying drawings under the inventive concept of the invention, or directly/indirectly application in other related technical fields are all included in the patent protection scope of the invention.

1. A vacuum sealing lid, applied to Mason jars and cooperating with a vacuum machine to realize vacuuming of Mason jars of different sizes, wherein the vacuum sealing lid comprises a first lid, a second lid, a first sealing ring and a second sealing ring,

the first lid is used to fit a first size of a Mason jar, a first placement groove is arranged on a top of the first lid, and a first vent hole is arranged in a middle of the first placement groove,

the second lid is used to fit a second size of a Mason jar, a second placement groove is arranged on a top of the second lid, and a second vent hole is arranged in a

middle of the second placement groove, the second lid is detachably connected with the first lid,

the first sealing ring is assembled on the first lid, the second lid and the first lid are sealed and connected through the first sealing ring to form a vacuum chamber, both the first vent hole and the second vent hole are connected to the vacuum chamber,

the second sealing ring is assembled on the second lid.

2. The vacuum sealing lid of claim 1, wherein the top of the first lid is recessed inward to form the first placement groove, and a top of the first placement groove is provided with a fillet or chamfer.

3. The vacuum sealing lid of claim 2, wherein a first boss is arranged in the middle of the first placement groove for positioning the vacuum machine, and the first vent hole is located on the first boss.

4. The vacuum sealing lid of claim 1, wherein the top of the second lid is protrusively provided with a shielding ring, so as to form the second placement groove, an inner side of a top of the second placement groove is provided with a fillet or chamfer.

5. The vacuum sealing lid of claim 4, wherein a second boss is arranged on the top of the second lid for positioning the vacuum machine, the second boss is located inside the shielding ring, the second vent hole is located on the second boss.

6. The vacuum sealing lid of claim 1, wherein a first annular positioning groove is arranged on one side of the first lid facing the second lid, the first sealing ring comprises a first clamping portion and a first sealing portion connected to each other, the first clamping portion is clamped in the first annular positioning groove, and the first sealing portion is tightly attached to the second lid or the first size of the Mason jar.

7. The vacuum sealing lid of claim 1, a second annular positioning groove is arranged on one side of the second lid facing the Mason jar, the second sealing ring comprises a second clamping portion and a second sealing portion connected to each other, the second clamping portion is clamped in the second annular positioning groove, and the second sealing portion is tightly attached to the second size of the Mason jar.

8. The vacuum sealing lid of claim 1, wherein the detachable connection between the first lid and the second lid is achieved through rotary buckles.

9. The vacuum sealing lid of claim 1, wherein both outer sides of the first lid and the second lid are provided with anti-slip grooves.

10. A vacuum sealing device, wherein the vacuum sealing device comprises the vacuum sealing lid and the vacuum machine,

the vacuum machine cooperates with either the first placement groove or the second placement groove to vacuumize the Mason jars.

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